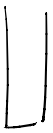
Seminar 7

luni, 22 noiembrie 2021 12:06

Automate Push-Down → LIC

$M = (K, \Sigma, \Gamma, \Delta, s, F)$

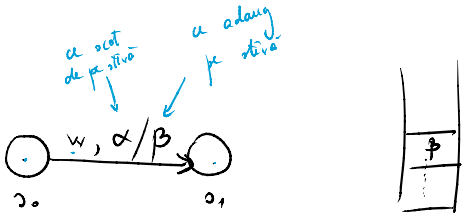
- K: mulțimea de stări
- Σ: alfabetul limbajului
- Γ: alfabetul stivei
- Δ: mulțimea de tranziții
- s: starea inițială
- F: mulțimea de stări finale



$\Gamma = \{\alpha, \beta\}$

$\Delta \subseteq (K \times \Sigma^* \times \Gamma^*) \times (K \times \Gamma^*)$

- K: starea din care plec
- Σ^*: șirul pe care fac tranziția
- Γ^*: ce scot de pe stivă
- K: starea în care ajung
- Γ^*: ce adaug pe stivă

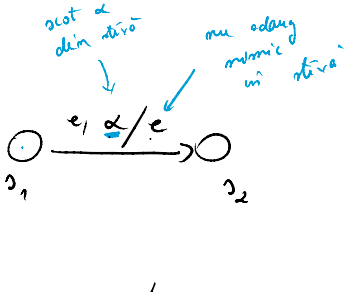


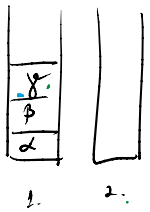
$s \in K$

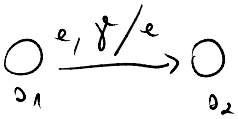
$F \subseteq K$

$|F| = 0$

$|K| \neq 0$







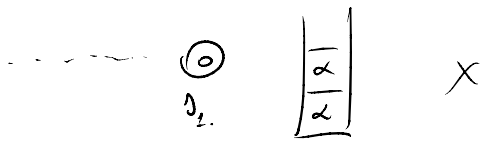


Un cuvânt e acceptat de APD dacă:

$(s, w, e) \vdash^* (f, e, e)$, $f \in F$

1. am șirul vid la intrare
2. am ajuns într-o stare finală

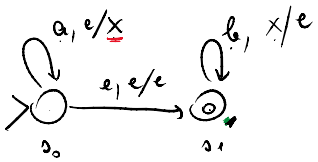
1. am simbol vid la intrare
2. sunt într-o stare finală
3. stiva este goală



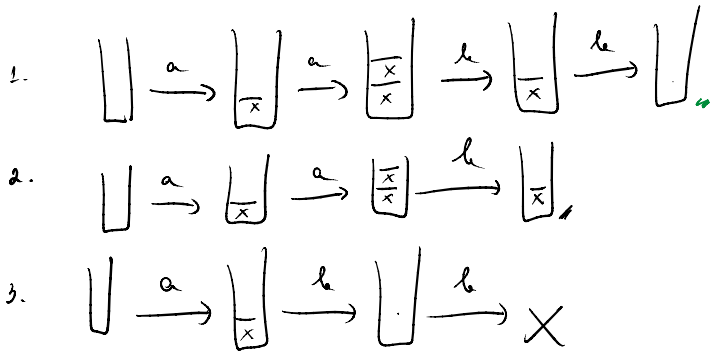
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1. $L = \{ a^m b^m / m \geq 0 \}$

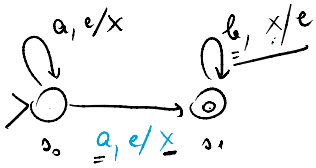
$S \to a S b \mid e$



1. a a b b ✓
2. a a b ✗
3. a b b ✗



2. $L = \{ e^m b^n / m \geq 1 \}$

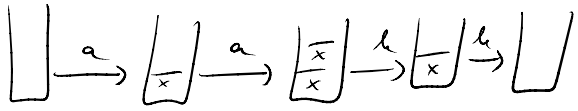


$= e$ ✓

$a a^3 b^3$

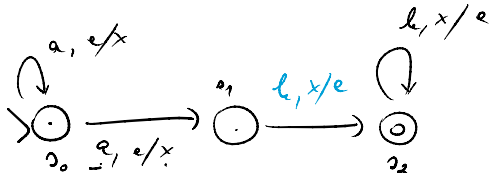
aabb ✓

e X

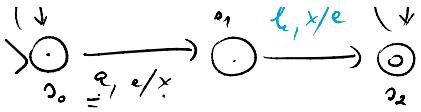


$a^m b^m$, $m > 1$.

ab ✗







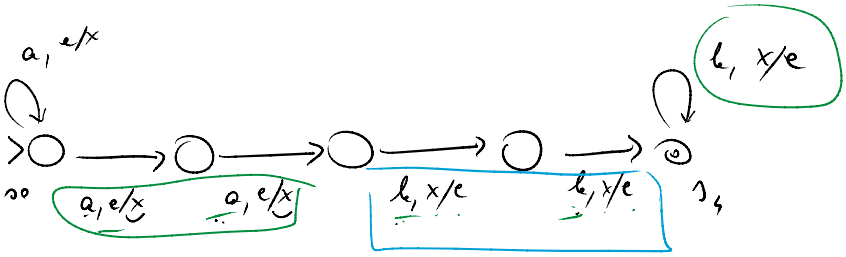


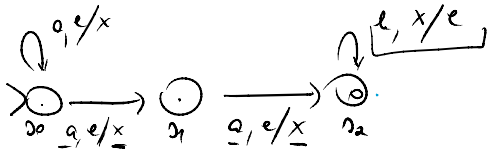
aabb



$a^n b^n$, $n \geq 1$

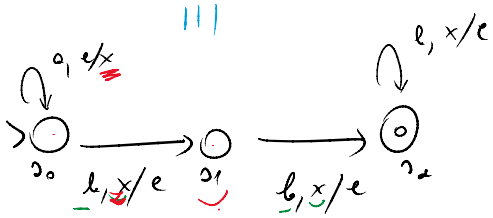
$n > 1$



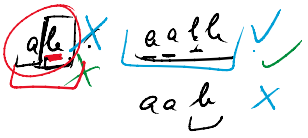




III









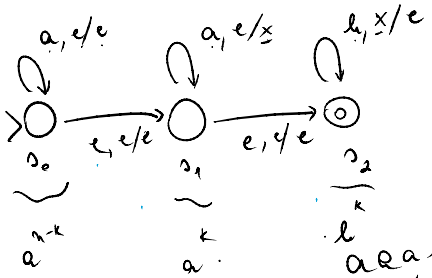
$a^n b^n$, $n \geq 2$

3. $L = \{ a^m b^k \mid m \geq k, \; m, k \geq 0 \}$

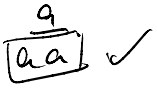
$a^n b^k$, $n, k \geq 0$.

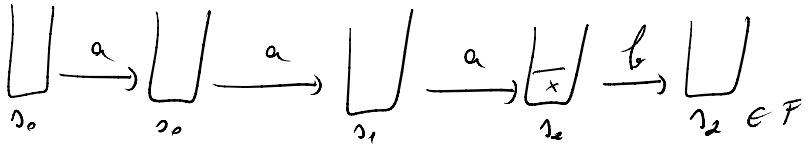
$a^{m-k} a^k b^k$

≥ 1 $k \geq 1$.



aaab





a

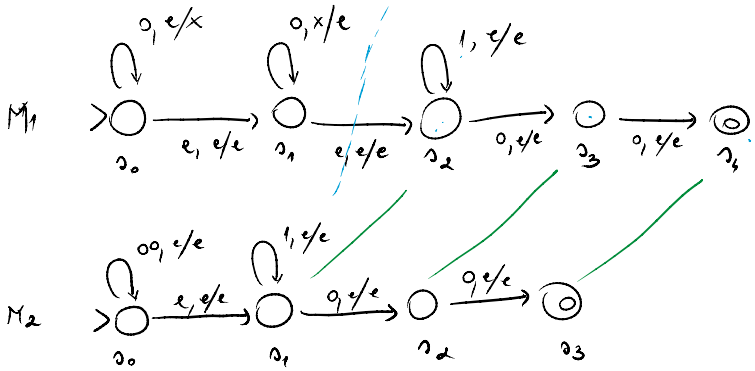
aa

$\forall L \in LR \Rightarrow L \in LIC$

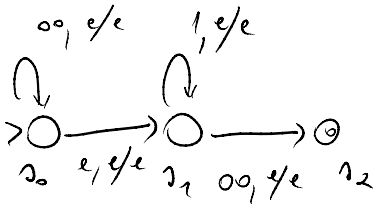
not $\forall L \in LIC \Rightarrow L \in LR$. gresit.



4. $L = \{ w_1 \in \Sigma^* / \; w_1$ e de forma: $(00)^n 1^k 0^2, \; n, k \geq 0 \}$.



M_3:

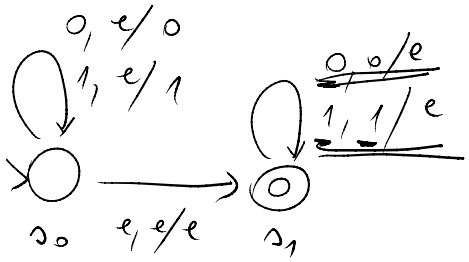


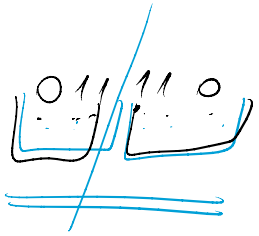
Care este nr. minim de stări pt. un AP3 care acceptă L?

a). 2 c). 4.

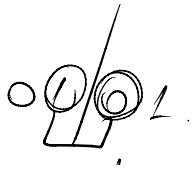
b). 3 d). 5

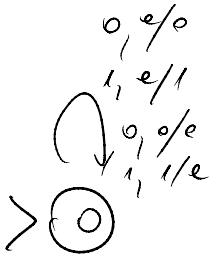
5. $L = \{ w_1 \in \Sigma^* / \; w_1$ e palindrom de lungime pară $w_1 = w_1' w_1'^R \}$ $\Sigma = \{0, 1\}$.

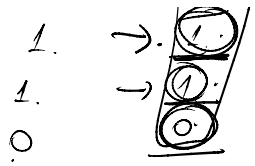


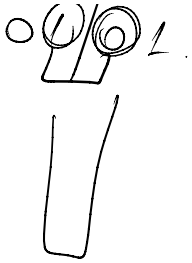










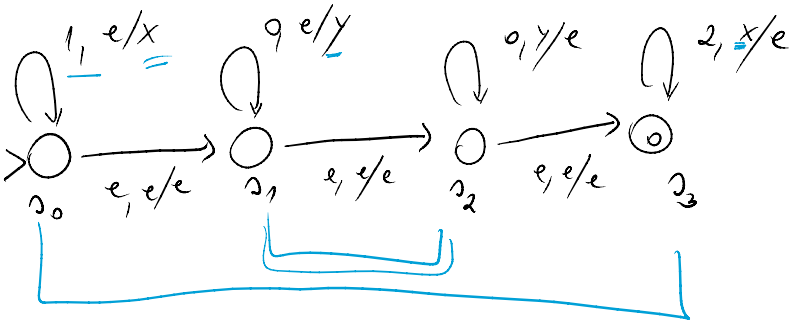


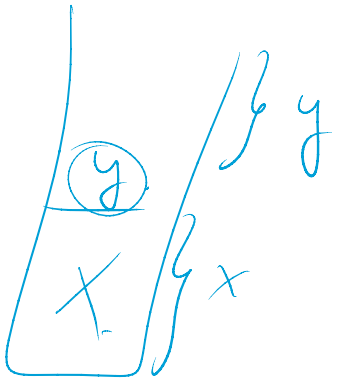


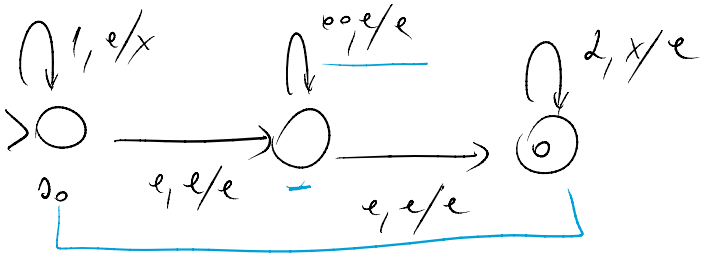


6. $L = \{ w \in \Sigma^* / w$ e de forma

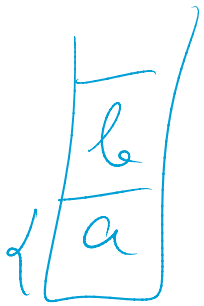
$1^k 0^m 0^n 2^k$, $n, k \geq 0$. $\}$







$a^n b^n c^n$ ~~$\in L/C$~~



$a^n b^k c^n d^k \notin L(G)$

